

# What is WebXR?

All specifications can be found here: [WebXR Device API](#)

The WebXR Device API is a kind of translator between a website and VR/AR hardware like headsets, motion sensors (that are also contained in smartphones) and hand controllers. Just as a normal website can "talk" to your mouse or keyboard, WebXR lets it "talk" to a VR headset too. This enables a browser to directly display a 3D world (VR) or overlay digital objects onto the real world (AR).<sup>1</sup>

This standard is being developed by the W3C (World Wide Web Consortium), which is the international standards organization that develops and maintains the web's core technical standards like HTML, CSS, etc. Website: [w3.org](https://www.w3.org)<sup>2</sup>

While it is still a work in progress, it has already been implemented in a couple of browsers and people have started to build web-based XR experiences with it. Before this, almost every VR/AR experience required its own installed app for a specific platform. WebXR lets developers build an experience once and run it directly in the browser without any installs, app store, or platform lock-in. The W3C's goal here is to enable websites to do everything a native AR/VR app can do, not just a weaker substitute version.

On its first announcement of the API's first public draft, the W3C states: "The ability to interact directly with immersive hardware is critical to ensuring that the web is well equipped to operate as a first-class citizen in this environment. ... The WebXR Device API provides the interfaces necessary to enable developers to build compelling, comfortable, and safe immersive applications on the web across a wide variety of hardware form factors."<sup>3</sup>



Fig. 1: The WebXR logo.<sup>4</sup>



Fig. 2: The W3C logo.<sup>5</sup>

---

<sup>1</sup> World Wide Web Consortium, WebXR Device API — Abstract (2026).

<sup>2</sup>World Wide Web Consortium (n.d.).

<sup>3</sup> World Wide Web Consortium, First Public Working Draft: WebXR Device API (2019).

<sup>4</sup>World Wide Web Consortium, Public Immersive Web Working Group Mailing List (2019).

<sup>5</sup>Wikipedia, World Wide Web Consortium (n.d.).

# Timeline of WebXR standard development

## The Past

### Web VR as the catalyst

#### Undated

Mozilla created a WebVR device API that made web based VR experiences possible. This was never ratified as a W3C standard, enabled by default in only a few browsers and supported only by a small number of devices. But still it showed the possibility and potential of browser based XR experiences. It has since been deprecated in favour of the WebXR Device API.<sup>6</sup>

#### 26 April 2018

Launch of "Hubs by Mozilla" as a browser-based social VR experiment, announced on the official Mozilla blog (one of the most well-known examples of web-based VR).<sup>7</sup>

**Interesting to know:** Mozilla ended its support for Mozilla Hubs in 2024. The codebase has since been maintained by the Hubs Foundation.<sup>8</sup>

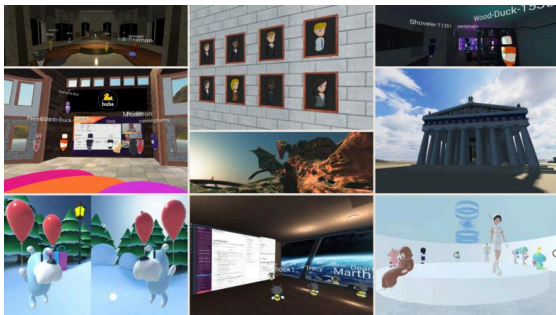


Fig. 3: Screenshots of Mozilla Hubs.<sup>9</sup>

### Beginning of Standardisation

<sup>6</sup> Mozilla Corporation, WebVR API, MDN Web Docs (n.d.).

<sup>7</sup> Mozilla Corporation, The Mozilla Blog, White (2018).

<sup>8</sup> Mozilla Corporation, End of Support for Mozilla Hubs (2024).

<sup>9</sup>Hubs Foundation (n.d.).

#### 24 September 2018

Start of the Immersive Web Working Group at the W3C (Initial Charter) with the goal of developing the WebXR Device API<sup>10</sup>

#### 5 February 2019

First Public Working Draft of the WebXR Device API.<sup>11</sup>

### First Browser implementation (Blink)

#### 10 December 2019

Google promotes Chrome 79 to the Stable channel which shipped the public working draft of the WebXR Device API for the first time in a Browser.<sup>12</sup>

#### 28 February 2020

Brave (Chromium-based) enables WebXR by default. (Previously a disabled feature.)<sup>13</sup>

#### 14 April 2020

Microsoft Edge 81 (Chromium-based) becomes available in the Stable channel on this day, according to Microsoft's own Edge blog; according to Microsoft's own compatibility table, the full WebXR Device API is documented from this version onward.<sup>14</sup>

<sup>10</sup> World Wide Web Consortium, Immersive Web Working Group Charter (2024).

<sup>11</sup> World Wide Web Consortium, WebXR Device API — Table of Contents (2026).

<sup>12</sup> Google LLC, Chrome Releases Blog & web.dev, Medley (2019/2020).

<sup>13</sup> Brave Software, GitHub Pull Request #4764 (2020).

<sup>14</sup> Microsoft Corporation, Microsoft Edge Blog & Microsoft Learn (2020/2021).

## Today

The W3C is still working on the WebXR Device API. As of September 2026, it is at the Candidate Recommendation Draft stage.<sup>15</sup> “Candidate Recommendation” is one of several maturity stages in the official W3C standardization process, just below the highest stage, “Recommendation.”<sup>16</sup> “Draft” denotes a sub-stage of the Candidate Recommendation stage.<sup>17</sup>

As seen in the table below, the WebXR device API is fully implemented in three different browsers (Chrome, Edge, Brave) that all run on Google's Blink engine and partially implemented in Apple's Safari browser (only VR Modules for VisionOS).

**Table 1: WebXR Status and Browser Market Share**

| Browser            | Engine           | WebXR-Status                       | EU Users 2026 |
|--------------------|------------------|------------------------------------|---------------|
| Chrome             | Blink (Chromium) | Full                               | 62.66 %       |
| Edge               | Blink (Chromium) | Full                               | 6.66 %        |
| Brave              | Blink (Chromium) | Full                               | -             |
| Safari             | WebKit           | partial (VR only, Vision Pro only) | 17.8 %        |
| Firefox            | Gecko            | none                               | 4.61 %        |
| Meta Quest Browser | Blink (Chromium) | Full                               | -             |
| PICO Browser       | Blink (Chromium) | Full                               | -             |

*Note. Own compilation. Europe Users figures are from StatCounter Global Stats, reflecting the most recent monthly snapshot available at the time this document was compiled (early September 2026); the underlying data month is August 2026. "Europe" is StatCounter's continental region definition and is not equivalent to the EU-27 specifically — no comparably reliable primary source offering an EU-only aggregate could be found. A dash (—) indicates no comparably reliable regional figure could be found.*

<sup>15</sup> World Wide Web Consortium, WebXR Device API — Status of This Document (2026).

<sup>16</sup> World Wide Web Consortium, Types of Documents W3C Publishes (n.d.).

<sup>17</sup> World Wide Web Consortium, Types of Documents W3C Publishes (n.d.).

## Outlook

No date for WebXR's overall advancement to final W3C Recommendation can be seriously predicted. For the core API itself, the Charter names no target date, instead explicitly describing it as a "living specification" with yearly updates rather than a fixed endpoint ("Expected W3C Recommendation: yearly basis (living specification)").<sup>18</sup>

The decisive criterion for the WebXR device API to advance to Recommended status is described as follows: at least two independent browser engines with a complete, verifiably interoperable implementation. Currently, no second candidate alongside Blink (Chrome/Edge/Brave) is apparent: Firefox (Gecko) does not implement WebXR at all, and according to the official W3C participant list of the Immersive Web Working Group, Mozilla is no longer represented there as an organization. Apple (WebKit/Safari) is actively involved as Working Group co-chair but that does not guarantee a complete implementation.<sup>19</sup>

### What Would Concretely Change Upon Advancing to Recommendation?

1. Patent Protection / Royalty-Free Use

"A W3C Recommendation has commitments from Working Group members to royalty-free licensing for implementations."<sup>20</sup>

That means that only upon advancing to Recommendation do all organizations participating in the Working Group make a legally binding commitment to make their patents on the relevant technology available royalty-free. Before that, this safeguard does not formally exist.

2. W3C states about the Recommendation stage: "W3C recommends the wide deployment of a Recommendation as a standard for the Web."<sup>21</sup>

Whether advancing to Recommendation would, in practice, actually trigger new browser implementations (e.g., whether Mozilla or Apple would then fully follow suit) cannot be substantiated by any of the primary sources reviewed.

---

<sup>18</sup> World Wide Web Consortium, Immersive Web Working Group Charter — Deliverables Table (2024).

<sup>19</sup> World Wide Web Consortium, Immersive Web Working Group — Participants (n.d.).

<sup>20</sup> World Wide Web Consortium, Types of Documents W3C Publishes (n.d.).

<sup>21</sup> World Wide Web Consortium, Types of Documents W3C Publishes (n.d.).

# WebAR Workarounds

As established previously in this course, the WebXR Device API's AR module does not currently work in Safari (desktop, iOS, or iPadOS). The following three tools all solve this problem the same way: they don't use the WebXR Device API at all. Instead, they run their own camera access, computer vision, and rendering directly in JavaScript/WebGL (which works on all browsers). All three tools differ in what AR capabilities they offer, the ownership behind them and their maintenance models.

## 8th Wall

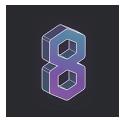


Fig. 5: The 8th Wall logo.<sup>22</sup>

**Documentation:** [8th Wall](#)

**Owned by:**

Niantic Spatial, Inc.

Since March 2026, its engine has been open source (MIT-licensed), but its SLAM (world-tracking) module remains closed-source.

**Maintained by:**

Niantic Spatial, Inc.

**Capabilities:**

SLAM/world tracking (closed binary only), Face Effects, Image Targets, Sky Effects.<sup>23</sup>

**How it works:**

8th Wall runs its own SLAM engine that does camera-based environment mapping.

---

<sup>22</sup>8th Wall / Niantic Spatial, Inc. (2026).

<sup>23</sup>8th Wall / Niantic Spatial, Inc. (2026).

## MindAR



Fig. 6: The MindAR logo.<sup>24</sup>

**Documentation:** [MindAR](#)

**Owned by:**

open source

**Maintained by:**

A single individual developer, going by "hiukim" on GitHub.<sup>25</sup>

**Capabilities:** Image and face tracking<sup>26</sup>

**How it works:**

Image tracking uses TensorFlow.js purely as a GPU compute engine (not for machine learning in the AI sense); face tracking is based on a face-landmark/mesh model.

---

<sup>24</sup>hiukim, GitHub, MindAR Documentation (n.d.).

<sup>25</sup>hiukim, GitHub — mind-ar-js README (n.d.).

<sup>26</sup>hiukim, GitHub (n.d.).

## AR.js



Fig. 7: The AR.js logo.<sup>27</sup>

**Documentation:** [AR.js](#)

**Owned by:**

open source

**Maintained by:**

The AR.js org  
It's a GitHub organization rather than a single person or a company.<sup>28</sup>

**Capabilities:** Marker tracking, location-based (GPS) AR, and image tracking (NFT).<sup>29</sup>

**How it works:**

AR.js runs marker recognition, GPS placement, and image tracking in plain JavaScript via WebGL and the device's camera/GPS APIs.

---

<sup>27</sup>AR-js-org, GitHub Pages, AR.js Documentation (n.d.).

<sup>28</sup>AR-js-org, GitHub — AR.js README (2026).

<sup>29</sup>AR-js-org, GitHub (2026).

Comparison Table

| Capability                            | WebXR (native)                     | 8th Wall                                               | MindAR               | AR.js                              |
|---------------------------------------|------------------------------------|--------------------------------------------------------|----------------------|------------------------------------|
| <b>Maintainer</b>                     | W3C (multi-vendor, browser makers) | Corporation (Niantic Spatial)                          | Individual developer | Community organization (AR.js org) |
| <b>License</b>                        | Open web standard                  | Mixed: MIT engine + closed-source SLAM binary          | MIT                  | MIT (+ LGPLv3 for artoolkit5-js)   |
| <b>World Tracking (SLAM)</b>          | Not part of the spec               | Yes (closed binary only)                               | No                   | No                                 |
| <b>Hit Test (surface placement)</b>   | Yes                                | Yes                                                    | No                   | No                                 |
| <b>Marker Tracking</b>                | No                                 | No                                                     | No                   | Yes                                |
| <b>Image Tracking</b>                 | No                                 | Yes                                                    | Yes                  | Yes (NFT)                          |
| <b>Face Tracking</b>                  | No                                 | Yes                                                    | Yes                  | No                                 |
| <b>Location-based (GPS) AR</b>        | No                                 | No                                                     | No                   | Yes                                |
| <b>Depth Sensing / Occlusion</b>      | Yes                                | Limited (mesh-based occlusion + sky segmentation only) | No                   | No                                 |
| <b>Anchors (persistent placement)</b> | Yes                                | No                                                     | No                   | No                                 |
| <b>Hand Input</b>                     | Yes                                | No                                                     | No                   | No                                 |

*Note: This table is a custom summary made by your teacher's own research. Proceed with caution and confirm functionalities yourself before basing your projects on it as this landscape is constantly changing.*

# Attachment

Sources: WebXR Browser Status Table ("Today" section)

Sources for the Engine and WebXR Status columns:

Chrome: Chrome Releases Blog and web.dev (Google).<sup>30</sup>

Edge: Microsoft Edge Blog and Microsoft Learn (Microsoft).<sup>31</sup>

Brave: brave-core Pull Request #4764 (Brave Software).<sup>32</sup>

Safari: WebKit Features in Safari 18.0 (Apple / WebKit Project).<sup>33</sup>

Firefox: WebXR Device API — MDN Web Docs (Mozilla).<sup>34</sup>

Meta Quest Browser: Meta Horizon OS Developers Blog (Meta Platforms).<sup>35</sup>

PICO Browser: Resources — PICO 4 Ultra (PICO Technology).<sup>36</sup>

Sources for the Europe Users column: StatCounter Global Stats.<sup>37</sup>

Supplementary explanation: Brave, Meta Quest Browser, and PICO Browser do not appear as separate line items in the StatCounter dataset. For Brave, there is a specific, verifiable technical reason: per Brave's own official documentation, "Brave uses the same User Agent request header as Chromium for web compatibility reasons" — meaning tools such as StatCounter, which identify browsers via the User Agent string, cannot distinguish Brave traffic from Chrome traffic, so Brave's usage is folded into Chrome's reported share.<sup>38</sup> For Meta Quest Browser and PICO Browser, no comparably reliable region-specific market-share source could be found at all; standalone-headset browsers are generally not tracked by mainstream browser-share tools.

## Sources

WebXR vs. Workarounds Comparison Table

Hit Test Module, shipping since Chrome/Edge 81 (2020).<sup>39</sup> Depth Sensing Module (shipping since Chrome/Edge 90, 2021) and Hand Input Module (shipping since Chrome 131, 2024).<sup>40</sup>

Anchor Module, shipping since Chrome/Edge 85 (2020).<sup>41</sup> Image and marker tracking are not part of any shipping WebXR module — the closest proposal, the "WebXR Marker Tracking Module," is only a Draft Community Group Report from July 2021.<sup>42</sup> Face tracking is confirmed absent by a direct statement on WebXR's own GitHub repository.<sup>43</sup>References

---

<sup>30</sup>Google LLC, Chrome Releases Blog (2019); web.dev (2020).

<sup>31</sup>Microsoft Corporation, Microsoft Edge Blog (2020); Microsoft Learn (2021).

<sup>32</sup>Brave Software, GitHub (2020).

<sup>33</sup>Apple Inc. / WebKit Project, WebKit Blog (2024).

<sup>34</sup>Mozilla Corporation (n.d.). WebXR Device API.

<sup>35</sup>Meta Platforms, Meta Horizon OS Developers Blog (2022).

<sup>36</sup>PICO Technology Co. Ltd., PICO Developer (n.d.).

<sup>37</sup>StatCounter (2026).

<sup>38</sup>Brave Software, GitHub Wiki (2023).

<sup>39</sup>World Wide Web Consortium (2025). WebXR Hit Test Module.

<sup>40</sup>web-platform-dx, Web Platform Features Explorer (n.d.). Depth Sensing; Hand Input.

<sup>41</sup>Immersive Web Working Group, W3C (2026). WebXR Anchor Module.

<sup>42</sup>Immersive Web Working Group, GitHub (2021). WebXR Marker Tracking Module.

<sup>43</sup>immersive-web/webxr, GitHub Issue #1150 (2020).

Citation style: Swiss/German academic footnote system. In-text superscript footnotes give a short citation (Author/Organization, Year[, locator]). Full bibliographic details for every source are listed once, alphabetically, in the References section at the end of this document, in the form: Author/Organization (Year). Title. Available at: URL (Accessed: Date).

8th Wall / Niantic Spatial, Inc. (2026). Goodbye 8thwall.com. Hello 8thwall.org. Available at: <https://8thwall.org/blog/8th-wall-open-source> (Accessed: 3 September 2026).

AR-js-org, GitHub (2026). AR.js – Image Tracking, Location Based AR, Marker Tracking. All on the Web. Available at: <https://github.com/AR-js-org/AR.js/> (Accessed: 3 September 2026).

AR-js-org, GitHub Pages, AR.js Documentation (n.d.). Available at:

<https://ar-js-org.github.io/AR.js-Docs/> (Accessed: 7 September 2026).

Apple Inc. / WebKit Project, WebKit Blog (2024). WebKit Features in Safari 18.0. Simmons, J., 16 September 2024, section "Spatial Web → WebXR". Available at:

<https://webkit.org/blog/15865/webkit-features-in-safari-18-0/> (Accessed: 2 September 2026).

Brave Software, GitHub (2020). Remove WebXR from disabled feature list · Pull Request #4764, brave/brave-core, merged 28 February 2020. Available at:

<https://github.com/brave/brave-core/pull/4764> (Accessed: 2 September 2026).

Brave Software, GitHub Wiki (2023). Detecting Brave (for Websites). Available at:

[https://github.com/brave/brave-browser/wiki/Detecting-Brave-\(for-Websites\)](https://github.com/brave/brave-browser/wiki/Detecting-Brave-(for-Websites)) (Accessed: 3 September 2026).

Google LLC, Chrome Releases Blog (2019). Stable Channel Update for Desktop, 10 December 2019. Available at:

<https://chromereleases.googleblog.com/2019/12/stable-channel-update-for-desktop.html>; and Google LLC, web.dev (2020). Augmented Reality: You May Already Know It. Medley, J., 13 February 2020. Available at: <https://web.dev/articles/web-ar> (Accessed: 2 September 2026, both sources).

hiukim, GitHub (n.d.). mind-ar-js – Web Augmented Reality: Image Tracking, Face Tracking.

Available at: <https://github.com/hiukim/mind-ar-js> (Accessed: 3 September 2026).

hiukim, GitHub, MindAR Documentation (n.d.). Available at:

<https://hiukim.github.io/mind-ar-js-doc/> (Accessed: 7 September 2026).

Hubs Foundation (n.d.). Welcome to Hubs. Available at: <https://docs.hubsfoundation.org/>

(Accessed: 5 September 2026).

immersive-web/webxr, GitHub Issue #1150 (2020). Does WebXR support ARCore

Augmented Faces for web-based AR? Available at:

<https://github.com/immersive-web/webxr/issues/1150> (Accessed: 3 September 2026).

Immersive Web Working Group, GitHub (2021). WebXR Marker Tracking Module (Draft Community Group Report, 2 July 2021). Available at:

<https://github.com/immersive-web/marker-tracking/blob/main/index.html> (Accessed: 3 September 2026).

Immersive Web Working Group, W3C (2026). WebXR Anchors Module. Available at:

<https://immersive-web.github.io/anchors/> (Accessed: 3 September 2026).

Meta Platforms, Meta Horizon OS Developers Blog (2022). Achieve Better Rendering and Performance with WebXR Layers in Oculus Browser and WebXR Web Launch.

Available at:

<https://developers.meta.com/horizon/blog/webxr-web-launch-mobile-matterport/> (Accessed: 2 September 2026).

Microsoft Corporation, Microsoft Edge Blog (2020). Update on Stable Channel Releases for Microsoft Edge, updated 14 April 2020. Available at: <https://blogs.windows.com/msedgedev/2020/03/20/update-stable-channel-releases/>; and Microsoft Corporation, Microsoft Learn (2021, last updated 2025). Using WebXR with Windows Mixed Reality. Available at: <https://learn.microsoft.com/en-us/windows/mixed-reality/develop/javascript/webxr-overview> (Accessed: 2 September 2026, both sources).

Mozilla Corporation (2024). End of Support for Mozilla Hubs. Available at: <https://hubs.mozilla.com/labs/sunset/> (Accessed: 2 September 2026).

Mozilla Corporation (n.d.). WebVR API. MDN Web Docs. Available at: [https://developer.mozilla.org/en-US/docs/Web/API/WebVR\\_API](https://developer.mozilla.org/en-US/docs/Web/API/WebVR_API) (Accessed: 2 September 2026).

Mozilla Corporation (n.d.). WebXR Device API. MDN Web Docs. Available at: [https://developer.mozilla.org/en-US/docs/Web/API/WebXR\\_Device\\_API](https://developer.mozilla.org/en-US/docs/Web/API/WebXR_Device_API) (Accessed: 2 September 2026).

Mozilla Corporation, The Mozilla Blog (2018). Enabling Social Experiences Using Mixed Reality and the Open Web. White, S., 26 April 2018. Available at: <https://blog.mozilla.org/en/mozilla/enabling-social-experiences-using-mixed-reality-and-the-open-web/> (Accessed: 2 September 2026).

PICO Technology Co. Ltd., PICO Developer (n.d.). Resources — PICO 4 Ultra. Available at: <https://developer.picoxr.com/pico4-ultra/> (Accessed: 3 September 2026).

StatCounter (2026). Browser Market Share Europe. Available at: <https://gs.statcounter.com/browser-market-share/all/europe> (Accessed: 3 September 2026).

web-platform-dx, Web Platform Features Explorer (n.d.). Depth Sensing; Hand Input. Available at: <https://web-platform-dx.github.io/web-features-explorer/features/webxr-depth-sensing>; <https://web-platform-dx.github.io/web-features-explorer/features/webxr-hand-input> (Accessed: 3 September 2026, both sources).

Wikipedia, World Wide Web Consortium (n.d.). Available at: [https://en.wikipedia.org/wiki/World\\_Wide\\_Web\\_Consortium](https://en.wikipedia.org/wiki/World_Wide_Web_Consortium) (Accessed: 5 September 2026).

World Wide Web Consortium (2019). First Public Working Draft: WebXR Device API. Available at: <https://www.w3.org/news/2019/first-public-working-draft-webxr-device-api/> (Accessed: 3 September 2026).

World Wide Web Consortium (2024). Immersive Web Working Group Charter (Charter History; Deliverables Table). Available at: <https://www.w3.org/2024/09/immersive-Web-wg-charter.html> (Accessed: 2 September 2026).

World Wide Web Consortium (2025). WebXR Hit Test Module. Available at: <https://www.w3.org/TR/webxr-hit-test-1/> (Accessed: 3 September 2026).

World Wide Web Consortium (2026). WebXR Device API — Candidate Recommendation Draft, 9 June 2026 (Abstract; Table of Contents; Status of This Document). Available at: <https://www.w3.org/TR/webxr/> (Accessed: 3 September 2026).

World Wide Web Consortium (n.d.). About us. Available at: <https://www.w3.org/about/> (Accessed: 4 September 2026).

- World Wide Web Consortium (n.d.). Immersive Web Working Group — Participants. Available at: <https://www.w3.org/groups/wg/immersive-web/participants> (Accessed: 2 September 2026).
- World Wide Web Consortium (n.d.). Types of Documents W3C Publishes. Available at: <https://www.w3.org/standards/types/> (Accessed: 2 September 2026).
- World Wide Web Consortium, Public Immersive Web Working Group Mailing List (2019). WebXR Device API Logo Proposal. Cannon, A. R., 10 July 2019. Available at: <https://lists.w3.org/Archives/Public/public-immersive-web-wg/2019Jul/0002.html> (Accessed: 12 September 2026).